Homework 2
FIN2020 Fall 2023

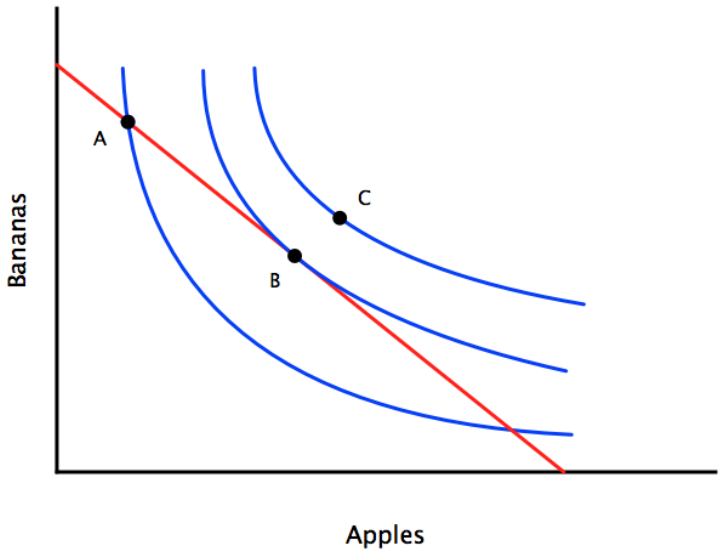
Due at **Midnight Tuesday, Oct 17 2023 Beijing Time**

Problem 1

The figure below illustrates consumer's indifference curve and budget constraint in a two-good economy of apples and bananas as we discussed in class. Let p_a and p_b denote the unit prices of apples and bananas. Suppose that consumer's utility function takes the form

$$\alpha u(c_a) + (1-\alpha)u(c_b),$$

where u is an increasing and concave function, c_a and c_b are the quantities of apples and bananas that consumed by the consumer, and parameter $\alpha \in (0,1)$ measures how much the consumer likes apples relative to bananas. The consumer's total income is $Y > 0$. Blue lines are indifference curves with different levels of utilities. The red line represents the budget constraint.



a. In the graph, there are three consumption bundles denoted by A, B and C. Which one is the optimal? Why.

b. Let c_a^* and c_b^* denote the number of apples and bananas included in the optimal bundle. Please write down two conditions that these quantities should satisfy.

c. Assume the utility function is a natural log function, i.e., $u(x) = \ln(x)$. Solve the optimal consumptions in terms of income Y.

Problem 2

Consider a consumer who receives income Y at the beginning of period $t = 0$, which he or she divides up into an amount c_0 to be consumed and an amount s to be saved. Suppose that the consumer receives no additional income in period $t = 1$, so that all of his or her consumption during that period has to be purchased with the savings and the interest earned on savings from period $t = 0$. Let r denote the interest rate.

a. First, write the budget constraints for the two single period $t = 0$ and $t = 1$. Then, combine the two single-period constraints to obtain the consumer's present-value, or "lifetime", budget constraint.

b. Suppose the consumer's preference over consumption during the two periods can be described by the utility function

$$\ln(c_0) + \beta \ln(c_1),$$

where $0 < \beta < 1$ represents how patient the consumer is. Solve this consumer's problem: choose optimal consumptions to maximize the utility function subject to the lifetime budget constraint.

c. Show how the consumer's optimal choice of consumption growth c_1^*/c_0^* depends on the discount factor β and the interest rate r.

Problem 3

Consider a two-date economy and a consumer with utility function over consumption at each period

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma},$$

where $\gamma > 0$. Define the intertemporal utility function as

$$v(c_1, c_2) = u(c_1) + u(c_2).$$

Show that the consumer will always prefer a smooth consumption stream to a more variable one with the same mean, i.e.,

$$v(\frac{c_1 + c_2}{2}, \frac{c_1 + c_2}{2}) > v(c_1, c_2).$$